

ORKG Properties Ontology Consolidated: LLM-Driven Refinement of Crowdsourced Knowledge for Machine-Actionability

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Abstract

In the era of Generative AI, Scientific Knowledge Graphs (SKGs) have become instrumental for grounding Retrieval-Augmented Generation (RAG), especially for mitigating hallucinations. As a prominent example, the Open Research Knowledge Graph (ORKG) serves as a foundational element of emerging research data infrastructures. It distinguishes itself by modeling not only metadata but also the semantic content of research publications as fine-grained triples. However, the quality and utility of these triples heavily depend on the predicates defined in the ORKG ontology. Currently, the ontology suffers from quality degradation inherent to uncontrolled crowdsourcing, such as widely duplicated properties, inconsistent naming conventions, and ambiguous semantics, thereby inhibiting machine-actionability.

In this paper, we introduce the Consolidated ORKG Properties Ontology (OPO-Consolidated), a consolidated schema designed to resolve these inconsistencies. We present a semi-automated workflow that combines Large Language Models (LLMs) with human engineering to systematically clean and restructure the existing property set. Our evaluation, utilizing an established Gold Standard dataset of 153 research papers, demonstrates that OPO-Consolidated substantially improves schema conciseness (by resolving synonymy and redundancy) and semantic consistency (by enforcing uniform naming conventions) while maintaining the valuable semantic coverage of the data. Furthermore, we show that OPO-Consolidated ensures backward compatibility with existing data, providing a seamless migration path while establishing a machine-actionable foundation for established ORKG comparisons and future downstream tasks.

CCS Concepts

• **Information systems** → **Ontologies**; **Digital libraries and archives**; • **Computing methodologies** → *Lexical semantics*.

Keywords

Scientific Knowledge Graphs, Large Language Models, Open Research Knowledge Graph, Ontology Engineering, Ontology Refinement

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1 Introduction

The rapid acceleration of scientific publishing has rendered traditional document-based retrieval increasingly inefficient, necessitating a paradigm shift towards structured Scientific Knowledge Graphs (SKGs) [8, 32]. Among these, the Open Research Knowledge Graph (ORKG) [2] occupies a unique position as a community-driven platform that models research contributions as fine-grained RDF triples, thereby laying the groundwork for next-generation applications such as automated literature comparison and grounded Retrieval-Augmented Generation (RAG).

However, the utility of such a crowdsourced SKG is heavily dependent on the quality of its underlying schema. While crowdsourcing enables rapid growth, it introduces significant noise: contributors frequently ignore modeling best practices, resulting in a proliferation of duplicate properties. Predicates such as “method” or “dataset” appear under dozens of distinct URIs, differing only by capitalization, grammatical number, or trivial lexical variations. This semantic heterogeneity creates a paradox: while the graph contains extensive knowledge, its fragmentation hinders effective retrieval, querying, and reuse.

Addressing this issue presents a unique challenge. Unlike entity linking, where established external references (e.g., Wikidata, DBpedia) exist, there is no comprehensive, domain-agnostic gold standard for scientific properties. Consequently, the solution cannot be a simple mapping to an external vocabulary but requires an intrinsic consolidation of the existing 13,000+ crowdsourced properties into a canonical ontology.

In this work, we present a framework to resolve this ambiguity. We propose a hybrid neuro-symbolic pipeline that utilizes the semantic reasoning capabilities of Large Language Models (LLMs) alongside vector-based clustering to clean and consolidate the



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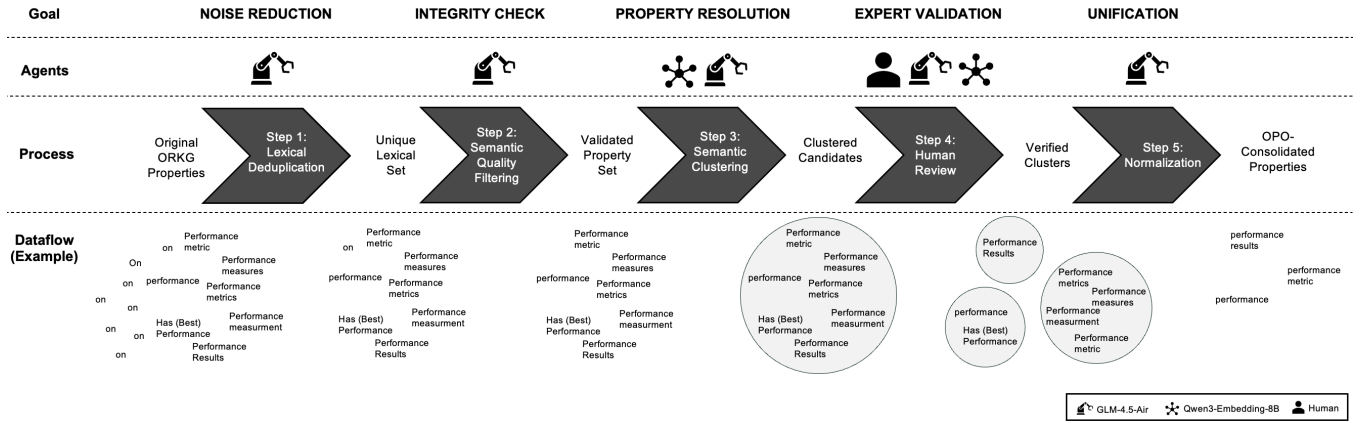


Figure 1: The OPO-Consolidated Workflow. Overview of the five-step neuro-symbolic pipeline transforming crowdsourced ORKG properties into a canonical ontology. The Dataflow Example traces the consolidation of the “performance” family: (1–2) Lexical duplicates and semantic noise (e.g., “on”) are filtered; (3) Embeddings cluster related terms; (4) Experts refine cluster granularity; (5) The LLM standardizes final labels. Note that the misspelling “performance measurment” in the figure is an original typo from the ORKG property set and is preserved intentionally as part of the example.

ORKG property landscape. A crucial feature of our approach is the preservation of data integrity: rather than deleting data, we implement a rigorous mapping strategy that ensures full backward compatibility for existing graph data. Our evaluation using an independent Gold Standard dataset of research properties [23] shows that the consolidated ontology highly significantly reduces semantic ambiguity in property search while preserving the integrity of the original graph structure. Our main contributions are:

- A neuro-symbolic pipeline combining LLMs and embeddings for property consolidation.
- A rigorous mapping strategy ensuring backward compatibility.
- A publicly available, significantly refined consolidated ontology (OPO-Consolidated) with reduced ambiguity.

To facilitate reproducibility and community adoption, we publicly release the complete pipeline, including source code, prompts, evaluation results, and log files, at <https://w3id.org/allmac/opo-consolidation>. The final OPO-Consolidated ontology is accessible via <https://w3id.org/orkg-properties-ontology-consolidated>. Documentation is available at <https://sandraschaftner.github.io/orkg-properties-ontology-consolidation>.

2 Background & Motivation

To understand the necessity of an automated consolidation pipeline, it is essential to analyze the unique position of the ORKG within the SKG landscape and the specific functional limitations caused by the current state of its ontology.

2.1 Lack of Alternatives for Scientific Properties

While numerous established ontologies exist for linking entities in scientific text – ranging from general knowledge bases like DBpedia and Wikidata to domain-specific taxonomies like the Computer Science Ontology (CSO) – there is a distinct lack of comprehensive ontologies for scientific properties (predicates). Alternative

approaches for SKG construction often limit themselves to a minimal set of predefined predicates (e.g., only seven properties in the CS-KG) [29]. However, capturing the nuance of scientific discourse requires a much broader vocabulary. The ORKG currently hosts approximately 13,000 properties accumulated over years of crowdsourcing. It is reasonable to assume that this set collectively covers the vast majority of necessary scientific relations. Therefore, rather than starting from scratch or limiting expressivity, the most viable path forward is to consolidate this extensive, albeit noisy, basis into a coherent standard.

2.2 Strategic Relevance of the ORKG

Refining the ORKG ontology is not merely a housekeeping task but a requirement for the broader research data ecosystem. The ORKG is widely referenced as a content-based SKG without domain limitations [5, 13, 17] and is prominently discussed in recent surveys on SKG construction [29] and the overarching vision of SKGs [32]. Critically, the ORKG is increasingly central to the German National Research Data Infrastructure (NFDI). It is integral to four consortia of the NFDI and is a candidate for the general KGI4NFDI project [14]. This institutional integration implies a long-term commitment to the platform, making the ontology standardization a prerequisite for future interoperability and large-scale adoption.

2.3 Functional Limitations

The current heterogeneity of properties creates three specific bottlenecks that hinder both human interaction and machine processing:

User Experience & Templates. When users annotate a paper, the system suggests existing properties via an autocomplete interface. Currently, searching for a common term like “dataset” yields a long list of semantically identical labels. This ambiguity confuses users, leading to arbitrary selections that further propagate inconsistency.

Machine Actionability. For automated agents, this fragmentation hinders both construction and retrieval. During SKG population, agents face the same selection dilemma as humans, struggling to choose one predicate among redundant variations. Moreover, executing a SPARQL query to retrieve all papers using a specific “method” is computationally expensive and error-prone if the notion “method” is distributed across dozens of property variations (e.g., *methods*, *used method*, *Method*).

Structured Comparisons. A core feature of the ORKG is the generation of research comparison tables [9]. These comparisons rely on aligning values under common predicates. If two papers describe the same property using different URIs, they cannot be merged into a single comparison row without significant manual effort, thereby defeating the purpose of a structured SKG.

3 Related Work

This work lies at the intersection of ontology engineering, quality assessment, and refinement, with a specific focus on crowdsourced ontologies. To the best of our knowledge, no comparable work currently addresses the automatic or semi-automatic consolidation of ontology properties within a single evolving schema. While some research exists on deduplication and consolidation, it predominantly focuses on the alignment and merging of multiple distinct ontologies [3, 15] rather than resolving internal semantic heterogeneity.

Quality Assessment. We adopt the property-centric criteria (Conciseness, Consistency, Semantic Accuracy) utilized by Mihindukulasooriya et al. [18], which align closely with the core characteristics identified in the systematic review by Wilson et al. [30].

Crowdsourced Ontologies. Community-based KGs notoriously suffer from the proliferation of duplicate URIs [19, 31]. Standard mitigation strategies include deduplication using similarity metrics and clustering [31]. We extend these approaches by integrating embeddings and LLMs to handle the semantic ambiguity typical of property labels.

LLMs in Ontology Engineering. Recent work highlights the utility of LLMs for ontology learning and engineering [4, 10, 16]. While they offer flexibility, purely automated systems often lack the precision required for authoritative schemas, requiring human intervention at carefully selected checkpoints [4, 10]. Consequently, we implement a human-in-the-loop strategy, where expert intervention is triggered for complex clusters, ensuring high precision where automated reasoning alone might fail.

4 Analysis of the ORKG Ontology

Before introducing our consolidation framework, it is crucial to quantify the extent of semantic heterogeneity within the ORKG. Our analysis of the ontology (as of December 2025) reveals severe fragmentation driven by uncontrolled crowdsourcing.

Lexical Redundancy. A purely lexical analysis shows that 11.5% of all properties are redundant duplicates caused merely by casing or trivial whitespace variations, leading up to 115 variations per property, each assigned a unique URI. This fragmentation forces



Figure 2: Automated Conformance Checking Protocol. The LLM utilizes guidelines derived from the ORKG Best Practices¹ to separate non-compliant labels (left) from canonical, compliant properties (right), ensuring consistent property standardization.

downstream applications to query massive disjunctions of URIs to retrieve complete data for a single property.

Semantic Noise and Modeling Patterns. Beyond simple duplicates, we observe structural quality issues that necessitate the rigorous filtering implemented in our prompts. Frequent patterns of low-quality data include:

- **Instance-as-Property Error:** A common modeling mistake is using specific instances as properties (e.g., *Linux*, *Windows*) instead of a generic relation like *operating system*.
- **Proliferation of Specific Properties:** Users often create highly specific properties instead of reusing generic ones, e.g., *Wind power calculated?* and *# Avg. Cols (target CEA)*.
- **Syntactic Noise:** The ontology contains numerous artifacts of uncuration data entry, including cryptic codes (e.g., *AG HG4III*, *TALIFF*), raw data values masquerading as predicates (e.g., *2020*, *70.3*), and grammatical fragments (e.g., *with*, *very significant*).

These findings underscore that simple string matching is insufficient for consolidation. A robust solution must be capable of distinguishing between valid scientific predicates and the significant noise inherent in crowdsourced data.

5 Methodology

This study introduces the OPO Consolidation Pipeline, a hybrid neuro-symbolic framework transforming the noisy ORKG property set into a canonical ontology. Prioritizing semantic precision and backward compatibility [31], we employ a rigorous mapping strategy where every consolidated property is linked to a canonical equivalent, ensuring no data loss.

We adopt the property-centric quality criteria of Mihindukulasooriya et al. [18] to address semantic heterogeneity. Excluding Syntactic Validity (which is guaranteed by our programmatic generation), we systematically address three dimensions:

Conciseness: Reducing redundancy via lexical deduplication (Step 1) and clustering (Step 3).

¹Source: https://orkg.org/help-center/article/42/Modeling_best_practices_for_resources_and_properties

Table 1: LLM Cluster Verification. Examples of broad embedding-based clusters ($\tau = 0.25$) and the subsequent LLM decision (KEEP or BREAK). Pure embeddings often merged distinct properties, which the LLM successfully separated. Conversely, the LLM correctly identified complex synonyms and kept them together.

Initial Cluster (Embeddings $\tau = 0.25$)	Semantic Analysis	LLM	Final Result (After $\tau = 0.15$ Refinement)
['C Statistic', 'Chi-Square', 'Goodness of fit']	Distinct Statistical Metrics	BREAK	Split into 3 properties: 1. <i>C Statistic</i> , 2. <i>Chi-Square</i> , 3. <i>Goodness of fit</i>
['has technique', 'main techniques used', 'Technique', 'Technique used', 'Techniques', 'Techniques/Methods', 'Underlying Technique', 'Uses technique']	Valid Synonyms	KEEP	Merged into single canonical property: <i>technique</i>
['has p-value', 'p-value', 't-value']	Distinct Variables	BREAK	Split into 2 groups: 1. <i>{p-value, has p-value}</i> , 2. <i>{t-value}</i>
['Error', 'Error Analysis', 'Error handling']	Process vs. Object	BREAK	Split into 3 distinct groups: 1. <i>Error</i> , 2. <i>Error Analysis</i> , 3. <i>Error handling</i>
['approach', 'approach', 'approach description', 'Approach name', 'Approach type', 'Approaches', 'approaches used', 'approach', 'covered approaches', 'Describes approach', 'developed approach', 'has_analysis_approach', 'type of approach']	Lexical Variations & Typos	KEEP	Merged into single canonical property: <i>approach</i>
['cohens kappa', 'Fleiss Score', 'Matthews Correlation Coefficient (MCC)']	Distinct Evaluation Metrics	BREAK	Split into 3 separate properties: 1. <i>cohens kappa</i> , 2. <i>Fleiss Score</i> , 3. <i>Matthews Correlation Coefficient</i>

Semantic Accuracy: Filtering out non-predicate labels (Step 2).

Consistency: Standardizing labels to a unified style guide (Step 5).

5.1 Model Selection

The pipeline operates through the interplay of a vector embedding model and a LLM.

Embedding Model. For vectorization, we used Qwen3-Embedding-8B². The selection was strictly driven by its top performance on the Massive Text Embedding Benchmark (MTEB) [22], standing as the leading open-weight model in the relevant categories of Semantic Textual Similarity (STS) and Clustering at the time of writing. Clustering Performance measures the model’s ability to separate distinct semantic domains (e.g., separating “political” properties from “physical” measurements). High performance here is a prerequisite to prevent creating clusters that merge unrelated terms. STS is the critical indicator for synonym detection (micro-clustering). As the core challenge involves identifying subtle variations of the same term, the model’s superior STS score ensures robust synonym identification.

LLM. GLM-4.5-Air³ was selected as the LLM. This decision was constrained by the available hardware resources (64 GB RAM). Within its size class, GLM-4.5-Air ranks among the top-performing open-weight models on the LMArena benchmark [7] at the time of writing. Technically, the model relies on an efficient Mixture-of-Experts (MoE) architecture [6, 12], which significantly reduces computational load by activating only a fraction of parameters during inference while maintaining state-of-the-art (SOTA) performance comparable to larger dense models. To ensure high-quality outputs, we employed advanced prompt engineering techniques, including few-shot learning with domain-specific examples and strict, concrete instructions [26, 28]. These strategies have been specifically evaluated as beneficial for ontology-related tasks in recent studies [10, 16, 27]. Furthermore, to ensure deterministic behavior and machine-readability, we set the sampling temperature to 0.0 and utilized Structured Outputs (JSON Schema) [25],

guaranteeing that the LLM’s responses are strictly parseable by the automation script.

5.2 The Consolidation Pipeline

The consolidation process consists of five sequential steps that progressively filter noise before performing advanced semantic clustering and normalization. Figure 1 provides a comprehensive overview of this workflow, illustrating the transformation of the “Performance” property family through the Dataflow tier.

Step 1: Lexical Deduplication. The initial stage focuses on reducing noise by resolving exact string matches and casing variations. In community-collected data, the same term often appears in multiple URIs with identical or trivially different labels.[18, 19, 31] Processing these as distinct entities in subsequent steps would be computationally wasteful. An example of such duplication patterns in the ORKG property set is provided in Section 4.

For each identified group of duplicates, the LLM is tasked with selecting the canonical URI. It evaluates the candidates based on adherence to Best Practices established by the ORKG, which guide users on how to properly define and label new graph elements (see Figure 2), specifically prioritizing lowercase labels and meaningful descriptions. Consequently, all duplicate URIs are mapped to the single “best” property within the group.

Step 2: Semantic Quality Filtering. Before attempting to cluster properties semantically, it is imperative to filter out noise labels. A common issue regarding semantic accuracy is the presence of properties that linguistically function as objects or topics rather than predicates. To address this, we implemented a strict Subject-Predicate-Object (SPO) verification using the LLM, serving as a fundamental integrity check for the ontology. For example, a label such as “communication technologies” is rejected because it describes a general research topic rather than a specific attribute. Furthermore, raw data values, overly specific variables, cryptic abbreviations, sentence fragments, and semantically void properties such as “on” (see Figure 1) are removed and mapped to the generic property `orkg:P41267` (“Link”). This strategy preserves the structural connection within the graph while neutralizing the semantic error of the original label.

²<https://huggingface.co/Qwen/Qwen3-Embedding-8B>

³<https://huggingface.co/unsloth/GLM-4.5-Air-GGUF>

Step 3: Semantic Clustering. This step represents the core of the semantic consolidation with the goal of property resolution, employing a two-phase strategy designed to balance two competing objectives: maximizing coverage (grouping all potential synonyms) and maintaining accuracy (avoiding false merges of distinct properties). Empirical experiments demonstrated that a single static threshold is insufficient, as the semantic distance between valid synonyms varies depending on the specificity of the domain.

In Phase 1, the system applies agglomerative clustering on the property embeddings using a loose distance threshold of $\tau = 0.25$ (cosine distance), determined through calibration experiments. While effective for capturing broad synonymy, this threshold occasionally merges distinct properties. To address this, Phase 2 leverages the LLM to validate each cluster. The model analyzes the semantic cohesion within the group and issues a **KEEP** or **BREAK** decision. If the LLM determines that a cluster contains distinct properties, it triggers a **BREAK** decision: the cluster is dissolved and immediately re-clustered using a strictly tighter threshold of $\tau = 0.15$, also determined through experimentation. We employ hard cluster assignment, meaning each property belongs to exactly one cluster. Furthermore, the re-clustering operates strictly on the members of the dissolved cluster, ensuring they are split into finer-grained subgroups without being reassigned to unrelated clusters. Table 1 provides concrete examples of these decisions, illustrating how the LLM successfully disentangles statistical metrics that share high vector similarity. This refinement allows the system to distinguish between closely related but ontologically distinct properties that the loose threshold initially merged.

Step 4: Human Review. While the hybrid automated approach is highly effective, certain domain-specific distinctions remain ambiguous for automated systems. Embeddings might group semantically related but ontologically distinct properties if they share a high degree of vector similarity. The Dataflow visualization (Figure 1, Step 4) illustrates such a case: the automated system initially aggregated varying performance aspects into a single broad cluster. To ensure ontological precision, an expert reviewer inspects these high-density clusters and, in this instance, architecturally refined the group into three distinct properties: *performance*, *performance metric*, and *performance results*. Following such an intervention, the system leverages vector embeddings to perform a localized re-clustering, assigning each property from the disputed group to the nearest of the newly defined centers.

Upon finalization of the cluster structure, the LLM selects the most appropriate canonical property for each cluster (from both Step 3 and Step 4), guided by the ORKG Best Practice rules (e.g., preference for concise, lowercase labels, see Figure 2). All other properties within the cluster are mapped to this canonical representative, completing the consolidation. Crucially, to ensure scientific reproducibility, all expert and LLM decisions (splits, merges, and canonical selections) are automatically serialized into structured log files, which are publicly available alongside the pipeline code.

Step 5: Normalization. The final step ensures that the surviving canonical properties adhere to a consistent style guide. This process is performed at the end of the pipeline to avoid the computational cost of normalizing properties that would otherwise be merged or filtered out in earlier steps. First, the LLM rewrites all remaining

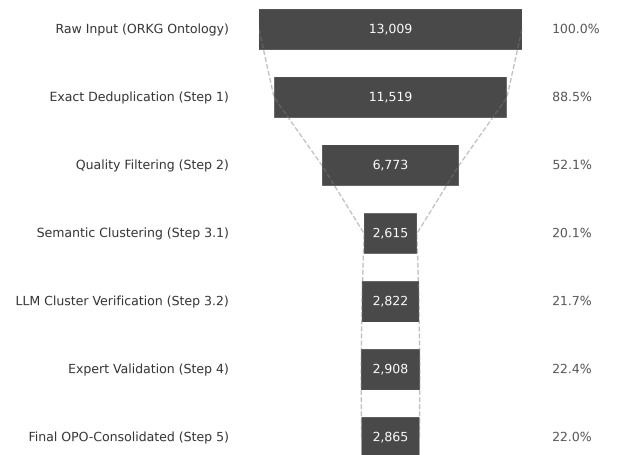


Figure 3: Consolidation Pipeline Funnel. The chart illustrates the reduction of property counts throughout the five processing steps. Note the significant compression during semantic clustering (Step 3.1) and the subsequent slight expansion during verification phases (Step 3.2 and Step 4), indicating the correction of over-merged clusters (false positives).

labels to strictly follow the conventions outlined in Figure 2, such as using lowercase, singular forms, and removing prefixes like “has” or “is”. For instance, as shown in the final Dataflow step (see Figure 1), “Performance metric” is standardized to “performance metric”. Crucially, this normalization process can introduce new collisions. For example, the distinct labels “used method” and “Methods” might both be normalized to “method”. To resolve these, a final LLM-based deduplication pass is triggered. The LLM is provided with the colliding candidates and instructs the selection of the single canonical URI based on the quality of the semantic description, explicitly prioritizing URIs with clear, academic definitions over those with empty or generic descriptions.

5.3 Mapping Strategy & Backward Compatibility

A critical requirement for the OPO Consolidation was to ensure that no data is lost during the transition. We utilized W3C standard predicates to define the relationship between the original (raw) properties and the consolidated (canonical) properties.

We employ `skos:exactMatch` [20] to link the original property to the canonical property. This indicates to search engines and users that the two terms are interchangeable in the context of information retrieval. Furthermore, we add `owl:equivalentProperty` [11] to provide formal logical equivalence, enabling reasoners to infer that a statement using the old property logically implies the same statement using the new property.

6 Evaluation Setup

To rigorously assess the performance and quality of the OPO Consolidation Pipeline, we adopted a mixed-methods evaluation strategy

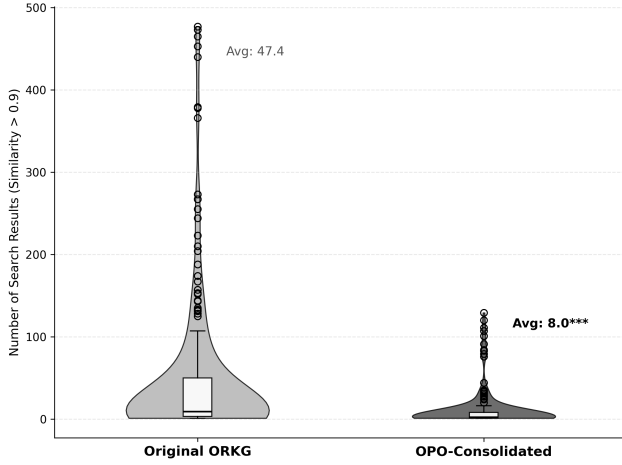


Figure 4: Distribution of Search Candidates ($\tau = 0.1$). The consolidation yields a statistically significant reduction in noise ($p < 0.001$, *), dropping the average candidate count from ~ 47 (Original, left) to a concise set of canonical properties (OPO-Consolidated, right).**

combining internal operational metrics with external ground-truth validation.

6.1 Operational Metrics and Gold Standard

The pipeline systematically logged operational metrics at each step, including property and clustering counts, LLM cluster validation decisions, human interventions, and generated mappings. All data were recorded in structured log files for quantitative analysis.

Complementing these internal metrics, we utilized the ORKG Gold Standard Property Dataset introduced by Nechakhin et al. [24] to verify the semantic integrity and usability of the resulting ontology. This dataset consists of 253 unique, human-annotated property labels extracted from 153 diverse research papers. Crucially, this dataset served as a strictly independent validation set, meaning these data points were not used during the pipeline’s development or configuration, allowing for an unbiased assessment of the system’s performance.

Using this external dataset, we tested two core hypotheses:

- (1) **Disambiguation (Usability):** The OPO-Consolidated significantly reduces semantic ambiguity compared to the raw ORKG property set.
- (2) **Data Preservation (Backward Compatibility):** The consolidation process preserves access to all original properties through valid mapping paths.

6.2 Metrics

6.2.1 Ambiguity Analysis. To quantify the improvement in usability, we measured semantic ambiguity in the vector space. We embedded all Gold Standard labels using Qwen3-Embedding-8B and performed a similarity search against both the original ORKG dataset and the OPO-Consolidated set, using a calibrated distance threshold of $\tau = 0.1$.

We define ambiguity as the average number of candidate properties returned for a specific search query. In the context of the ORKG, whether for human users or machine agents, this metric represents the number of properties returned for a label query when considering synonyms. An ambiguity score of 10 implies that a search yields 10 distinct property IDs. A higher number significantly increases the cognitive load and complexity for both humans and machines to select the correct property. Crucially, this ambiguity directly undermines machine actionability. An excessive selection of synonymous properties increases the likelihood that different URIs are assigned to the same semantic notion (e.g., by different agents or users), thereby rendering automated reasoning ineffective.

The threshold of $\tau = 0.1$ was selected based on empirical calibration. A looser threshold, such as $\tau = 0.15$, proved inappropriate for search retrieval, as it yielded an average of over 800 candidates for the original ontology and over 100 for the consolidated set. This confirms that while loose thresholds may serve clustering steps, strict thresholds are necessary for precise retrieval evaluation.

6.2.2 Provenance Path Validation. A primary concern in ontology consolidation is the potential loss of information. To test this, we performed a Provenance Path Validation. For every property in the Gold Standard, we identified its corresponding URI in the original ORKG dataset and traced its mapping path through the consolidated ontology using `skos:exactMatch` and `owl:equivalentProperty` assertions to verify that a valid canonical target exists.

7 Results and Discussion

The evaluation of the OPO Consolidation Pipeline demonstrates a decisive improvement in ontology quality without compromising historical data integrity. Our analysis confirms a statistically highly significant reduction in semantic ambiguity alongside 100% backward compatibility for legacy properties.

7.1 Quantitative Reduction

The consolidation pipeline successfully transformed the raw input of 13,009 properties into a curated ontology of 2,865 canonical properties (see Figure 3). The most significant compression occurred during the Semantic Clustering (Step 3), driven by the embedding-based detection of synonyms. Crucially, the reduction process was not strictly monotonic. After reaching the pipeline’s narrowest point of 2,615 candidate clusters (Step 3.1), the property count slightly increased to 2,822 following the LLM verification and further to 2,908 during the expert review (Step 4). This stabilization phase demonstrates the system’s hybrid safeguards: rather than accepting the most aggressive compression, the pipeline dissolved overly broad clusters (BREAK decisions) to recover distinct properties that had been falsely merged.

However, reduction was not the sole metric. The pipeline generated 10,120 semantic mappings, preserving the link between the raw data and the canonical schema. Notably, the sum of canonical properties and mappings does not exactly equal the raw input count of 13,009. The discrepancy of 24 properties is attributed to exact data record duplicates (identical URIs and metadata found in the raw export), for which no mapping was required. This confirms that every distinct entity removed from the active ontology was successfully captured in the mapping layer.

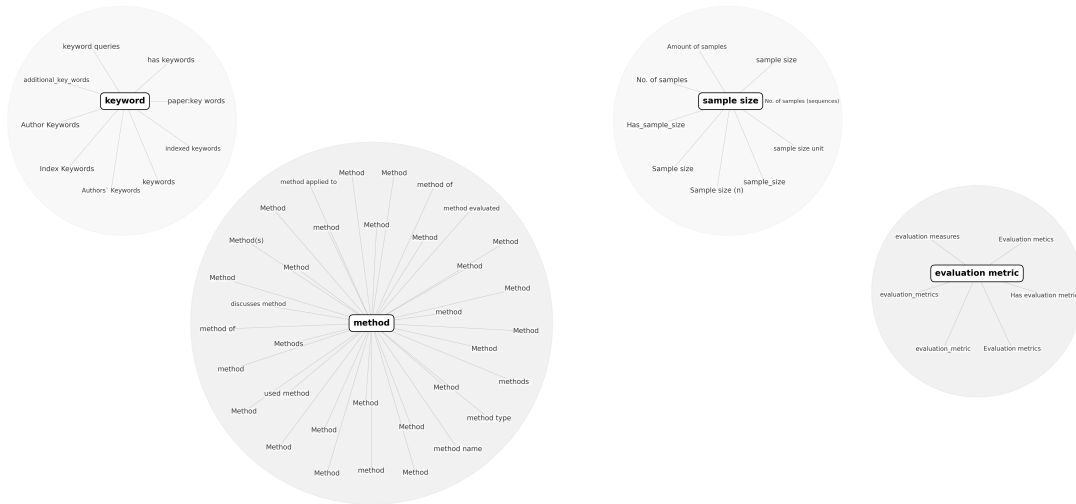


Figure 5: Visualizing consolidation: Bold central nodes represent canonical properties, merging redundant satellite properties from the original ORKG. The clusters illustrate the resolution of lexical variations, typos, and synonyms.

7.2 Ambiguity Reduction and Backward Compatibility

The evaluation using the Gold Standard demonstrates a massive improvement in usability. While the Original ORKG returned on average 47.4 candidates per query (similarity > 0.9) – often characterized by extreme redundancy such as 465 variations for “is about” – OPO-Consolidated dropped this average to 8.0. This represents a statistically significant ($p < 0.001$) 6-fold reduction in semantic noise. As illustrated in Figure 4, the search distribution shifted dramatically from a redundant “long tail” to a compact set of legitimate semantic neighbors, allowing for precise selection. The remaining outliers (> 100 candidates) correspond to high-density semantic regions. Specifically, they cluster around quantitative metrics (e.g., “number of papers”, “number of training data”) and generic relations (e.g., “operates on”), where the pipeline’s verification step correctly preserved distinct properties to maintain semantic precision rather than merging them into broad generalizations.

The Provenance Path Validation confirms a 100% backward compatibility with the original ORKG ontology. Every single property from the Gold Standard that existed in the original dataset could be successfully resolved to an OPO-Consolidated property via the ‘skos:exactMatch’ and ‘owl:equivalentProperty’ semantic links. This ensures that historical data remains fully accessible despite the schema consolidation.

7.3 Qualitative Analysis & Human Engineering

The hybrid neuro-symbolic architecture proved essential for achieving high precision. Figure 5 illustrates the qualitative impact of this consolidation. By analyzing individual clusters such as *method* or *evaluation metric*, we observe that the pipeline successfully resolves extensive noise — ranging from grammatical variations (singular/plural) to severe typos (e.g., “Evaluation metics”) and complex synonyms (e.g., “Amount of samples”) — into clean canonical representatives. This high output quality is attributable to strategic

intermediate verification steps: specifically, the LLM acts as a semantic verifier in Step 3, preventing the over-merging of properties that are mathematically close but ontologically distinct (as demonstrated in Table 1). Furthermore, the expert validation in Step 4 ensured precision in edge cases, requiring 52 manual architectural decisions to refine the final cluster definitions.

While the pipeline operates autonomously during execution, its success is fundamentally driven by human engineering. The contribution of this work lies in designing the cognitive architecture of the system. The pipeline’s performance is the result of over 30 refinement cycles, in which thresholds were empirically adjusted, LLM instructions were iteratively concretized, and few-shot examples were meticulously curated. We posit that current LLMs, despite their advanced capabilities, lack the agency to design such a sophisticated pipeline from scratch. However, the artifacts created by this work can serve as a blueprint for future agentic systems.

8 Learnings and Future Work

Consolidating crowdsourced ontologies is inherently challenging, requiring the resolution of both lexical and semantic ambiguities. Our hybrid pipeline addresses this through extensive human engineering of thresholds and prompts. Despite reliable performance, inherent trade-offs remain, with minor cases of over- or under-merging. While further refinement stages are possible, the evaluation demonstrates statistically highly significant improvements over the original ontology. We argue that OPO-Consolidated provides a superior baseline for subsequent expert curation or crowdsourced refinement [1, 21] compared to the raw data.

Although qualitative analysis confirms the effectiveness of the chosen embedding and LLM configuration, a systematic model benchmark could offer further insights. Currently, we prioritize refining the process logic over model variation.

Regarding future work, we recommend integrating deduplication directly into SKG construction to prevent redundancy at the

source [31] and enforce Best Practices. Finally, transferring this generic pipeline currently necessitates repeating the human engineering effort, specifically the manual calibration of clustering thresholds and the curation of domain-specific prompt examples. We are currently exploring agentic frameworks where LLMs autonomously refine these parameters, automating the adaptation to diverse ontologies.

9 Conclusion

This paper addressed the problem of semantic and lexical heterogeneity in the ORKG properties ontology, a core bottleneck for making the ORKG truly machine-actionable at scale. Starting from a noisy, crowdsourced property set of 13,009 predicates, we designed and implemented the OPO Consolidation Pipeline, a hybrid neuro-symbolic workflow that combines SOTA embeddings, LLM-based reasoning, and targeted expert validation to derive a canonicalized ontology with explicit backward-compatible mappings.

Our analysis shows that the pipeline substantially improves ontology quality along several dimensions. Quantitatively, it reduces the active property vocabulary to 2,865 canonical predicates while generating over 10,000 semantic mappings that preserve access to legacy URIs. From a usability perspective, the ambiguity analysis reveals a highly significant reduction in search-space noise, with every Gold Standard property retaining a valid mapping path to a canonical target. Qualitative inspection of high-frequency clusters further confirms that the pipeline resolves extensive redundancy, typos, and inconsistent modeling patterns into clean, reusable properties that align with ORKG Best Practices.

Beyond the ORKG, the proposed five-step consolidation workflow illustrates how LLMs and embeddings can be orchestrated into a practical, human-in-the-loop framework for ontology refinement. The public release of OPO-Consolidated, together with the full pipeline implementation, logs, prompts, and evaluation artifacts, provides an immediately usable resource for SKG applications and a reusable blueprint for consolidating other crowdsourced or weakly curated ontologies.

Declaration on Generative AI

During the preparation of this work, the authors utilized Generative AI for translation, rewording, improving style, and condensing the text. Furthermore, AI assistance was used to generate Python code for plotting data. The authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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